AI Analytics Copilot

Executive Report: winequality-red

1599 rows, 12 columns, file type: CSV

Executive Summary

The dataset comprises 1599 rows and 12 columns, all complete with no missing values. However, it contains 240 duplicate rows and 80 detected anomalies, indicating potential data quality issues. Several columns, notably 'residual sugar' (9.69%) and 'chlorides' (7.0%), exhibit a significant number of outliers.

Key Findings

- All 12 columns are fully populated with 0.0% null values.
- A substantial number of duplicate rows (240) are present in the dataset.
- Anomaly detection identified 80 anomalous rows, accounting for 5% of the total dataset.
- The 'residual sugar' column has the highest percentage of outliers at 9.69% (155 rows), followed by 'chlorides' at 7.0% (112 rows).
- Strong correlations exist, such as 'fixed acidity' with 'pH' (-0.683) and 'citric acid' (0.672).

Risks

- The presence of 240 duplicate rows could lead to biased statistical analysis or model training if not properly addressed.
- Significant outliers in columns like 'residual sugar' (9.69%) and 'chlorides' (7.0%) may distort data distributions and impact model robustness.
- The 80 detected anomalies suggest unusual data points that might represent errors or rare events requiring further investigation.

Opportunities

- Investigate the nature of the 240 duplicate rows to determine if they are valid repeated measurements or data entry errors.
- Analyze the characteristics of the 80 anomalous rows to understand their root causes and potential impact on insights.
- Leverage strong correlations, such as between 'fixed acidity' and 'pH' (-0.683), for potential feature engineering or to simplify model complexity.

Recommendations

- Address the 240 duplicate rows by either removing them or understanding their origin to ensure data integrity for analysis.
- Further investigate the 80 anomalous rows, particularly those with extreme values like 'residual sugar' at 15.4 or 'total sulfur dioxide' at 289.0, to determine their validity.
- Implement appropriate outlier handling strategies for columns with high outlier percentages, such as 'residual sugar' (9.69%) and 'chlorides' (7.0%), prior to any modeling efforts.

Data Quality Profile

Metric	Value
Duplicate rows	240
Total missing cells	0
Columns with outliers	12
Strong correlations found	4

Anomaly Detection

Method: dbscan. Flagged 1197 anomalous rows out of 1599.

Forecast

No forecast has been run for this dataset yet.